

CULIACAN, Mexico

WMO: 764120

Lat: 24.6350N

Lon: 107.4411W

Elev: 128

StdP: 14.63

Time Zone: -7.00 (MEC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.3	51.7	39.8	36.2	65.6	43.3	41.5	66.7	17.9	74.3	13.6	78.4	1.4	220	0.338

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	100.5	78.9	98.8	78.6	97.2	78.3	83.8	94.0	82.7	93.0	82.0	92.1	8.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.5	164.4	87.6	80.5	159.2	87.0	79.2	152.2	86.4	47.9	93.7	46.7	92.9	45.8	92.6	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.4	13.4	11.8	44.9	105.0	2.8	0.9	42.9	105.7	41.2	106.2	39.6	106.8	37.6	107.5		
			42.7	87.1	2.6	2.2	40.9	88.6	39.4	89.9	38.0	91.1	36.1	92.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	78.7	68.3	69.5	72.2	75.6	80.7	86.3	87.3	86.5	85.9	84.4	76.9	70.2	(6)
(7)		DBStd	7.79	4.27	3.50	3.22	2.68	3.13	2.82	2.42	2.47	2.52	3.24	4.13	3.90	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	30	19	4	1	0	0	0	0	0	0	0	0	6	(9)
(10)		CDD50	10476	568	547	687	768	952	1090	1157	1130	1078	1067	806	626	(10)
(11)		CDD65	5032	122	132	223	318	487	640	692	665	628	602	356	167	(11)
(12)		CDH74	60200	1351	1459	2619	3703	5886	8180	9065	8380	7713	6924	3372	1548	(12)
(13)		CDH80	30587	406	480	1120	1884	3325	4409	4966	4350	3882	3630	1639	496	(13)
(14)	Wind	WSAvg	3.8	2.9	3.3	3.8	4.5	4.4	5.0	4.5	3.7	3.6	3.4	3.0	3.0	(14)
(15)	Precipitation	PrecAvg	27.8	0.7	0.5	0.1	0.1	0.0	1.2	6.8	7.5	6.1	2.0	0.7	0.8	(15)
(16)		PrecMax	41.1	8.8	5.5	1.0	2.7	0.3	4.8	14.5	12.5	12.5	8.0	2.7	5.2	(16)
(17)		PrecMin	16.5	0.0	0.0	0.0	0.0	0.0	0.0	2.4	1.9	2.5	0.0	0.0	0.0	(17)
(18)		PrecStd	6.7	1.8	1.1	0.2	0.5	0.1	1.3	2.5	2.4	2.5	2.0	0.9	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.9	91.3	95.0	98.2	100.8	101.9	102.3	100.8	100.1	100.8	97.1	89.8	(19)
(20)			MCWB	68.7	69.4	69.6	71.1	73.9	79.2	80.5	81.9	82.5	78.2	72.2	69.0	(20)
(21)		2%	DB	86.4	88.0	91.2	94.6	98.6	99.0	99.0	98.6	97.5	98.6	94.7	87.6	(21)
(22)			MCWB	67.2	68.7	69.0	70.1	73.8	78.5	79.9	81.5	81.5	77.4	72.4	68.3	(22)
(23)		5%	DB	84.4	85.8	89.2	91.8	96.1	97.2	98.2	96.6	95.4	96.6	91.8	85.6	(23)
(24)			MCWB	67.1	67.8	68.8	69.4	73.1	77.9	79.7	80.7	80.9	78.1	72.4	68.3	(24)
(25)		10%	DB	82.3	82.8	86.3	89.8	93.6	95.3	95.9	94.8	93.6	94.7	89.7	82.7	(25)
(26)			MCWB	66.0	67.0	68.2	69.0	72.5	77.4	79.2	80.7	80.6	77.6	72.1	67.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.0	75.2	76.0	76.0	80.6	83.1	84.4	85.0	85.7	84.5	78.9	75.0	(27)
(28)			MCDWB	83.9	84.7	87.2	90.7	95.9	96.6	95.4	95.7	96.0	93.9	90.7	84.8	(28)
(29)		2%	WB	70.1	71.6	73.3	73.8	77.3	81.0	82.6	83.4	83.7	82.3	76.7	72.0	(29)
(30)			MCDWB	82.7	84.2	86.0	89.3	92.3	95.1	94.5	93.4	93.4	91.8	90.0	82.4	(30)
(31)		5%	WB	68.6	69.7	71.1	72.2	76.3	80.2	81.0	82.5	82.6	80.9	75.1	70.1	(31)
(32)			MCDWB	81.3	81.8	84.7	87.2	90.8	94.0	92.8	92.4	92.3	90.4	87.4	80.7	(32)
(33)		10%	WB	67.4	68.1	69.5	70.8	74.9	78.8	80.4	81.5	81.4	79.8	73.4	68.6	(33)
(34)			MCDWB	79.6	80.8	83.6	85.1	88.9	91.5	92.1	91.0	90.4	89.2	85.2	79.4	(34)
(35)	Mean Daily Temperature Range	MDBR	28.2	28.2	30.9	30.8	28.3	20.0	19.3	18.3	18.0	22.4	26.5	26.9	(35)	
(36)		5% DB	MCDBR	30.8	31.8	34.3	33.9	31.4	22.0	21.2	20.4	19.8	24.2	28.3	29.5	(36)
(37)			MCWBR	15.1	16.0	16.2	14.6	12.5	8.7	8.0	7.8	7.9	9.4	11.9	14.6	(37)
(38)		5% WB	MCDBR	27.5	28.4	29.7	29.8	26.4	20.6	19.6	18.8	18.6	20.8	26.0	26.0	(38)
(39)			MCWBR	14.4	15.3	16.1	14.6	12.6	9.1	8.2	8.3	8.5	9.1	12.6	14.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.304	0.307	0.337	0.374	0.419	0.425	0.406	0.409	0.350	0.312	0.301	(40)	
(41)		taud	2.647	2.607	2.568	2.466	2.403	2.360	2.400	2.460	2.417	2.616	2.681	2.650	(41)	
(42)		Ebn at Noon	308	313	315	305	291	276	274	280	278	293	301	303	(42)	
(43)		Edh at Noon	26	29	32	36	38	40	38	36	36	29	25	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1362	1659	2052	2332	2463	2286	2046	1952	1764	1722	1486	1296	(44)	
(45)		RadStd	109	129	73	120	100	97	92	68	73	85	79	63	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page